

GMD2

Lecture 07: Project Work

Midterm Evaluation

- Hybrid-undervisning (☹_☹)
- Det går (for) hurtigt, og der er meget indhold til hver gang
- Faglig niveau for højt/tilpas 50/50
- Fag på 2. semester en smule fragmenteret
- Generel tilfredshed med formatet (minus hybrid)
- God trivsel i klassen

Course Project

The goal of the project is to create a simple 2D game that uses best practices for organizing and architecting a game project. That is, the project must include gameplay systems and design elements solved through patterns and principles taught in the course.

The game project is used as reference and talking point at the exam

Project Work

- What are the groups?
- Have you decided on a game?
 - You don't need to know all the details before you start!
 - Iterate! Experiment!

Version Control

The source code of the project must be versioned using the git version control system and hosted on GitHub. By the final course session, the root of the repository must contain a README.md file containing key information on your project:

- A title and a brief description of the project
- A "how to play"-section (i.e. input, actions, game modes, etc..)
- A brief video demonstration of the project (hosted on YouTube)
- A link to a downloadable, playable Windows build of the project (hosted on itch.io)

Setup Version Control

- Use 15min on setting up the source code for the project
 - MonoGame template
 - .gitignore
 - Online GitHub remote
 - Add collaborators
- Upload repository link to itslearning

Reference & Value Types

Value Types

Structures

- Allocated (mostly) on stack
- Memory allocation done automatically
- Immediately removed when out of scope

int, bool, float, double, decimal, char, byte, short, long, enums, Custom structures

Reference Types

Classes

- You need to allocate memory
- Memory allocated on heap
- Garbage collected by CLR

Arrays, Strings, Custom classes

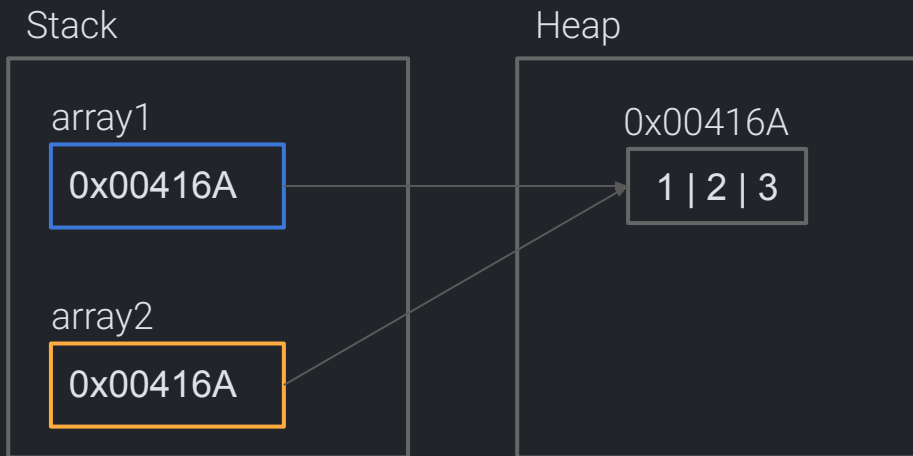
Reference & Value Types

```
var anotherObject = someObject;
```

Value Types



Reference Types



memory address is copied, not the actual value

Reference & Value Types

```
var a = 10;  
var b = a;  
b++;
```

Is a 10 or 11?

Reference & Value Types

```
int[] array1 = [1,2,3];  
int[] array2 = array1;  
array2[0] = 0;
```

Is array1[0] 0 or 1?

